

COMPENSATED VALUATION COMPLETIONS AT CONTENT-ONE AND PRIME-NORM POINTS, AND THE RAMIFICATION COST OF NORM-POWER NORMALIZATION

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ABSTRACT. We study the correlation of high prime multiplicity, low-depth credit and imbalance in the project's *abc* program. An elementary completion theorem prescribes exponents on both large arms of $1 + b = c$, forces norm-exponent content one, and proves an exact compensating radical formula. A second completion theorem uses a prime-element estimate with its Siegel-zero term retained to produce rational-prime auxiliary norms, moving valuation patterns, and squarefree complementary support. Its constants are uniform in a stated pseudopolynomial-modulus range, with at most one exceptional conductor explicitly excluded. Growing primes and growing prescribed depth can occur in these fully compensated completions. Separately, a sharp Jacobian ramification argument bounds the boundary degree of any rational map forcing an exact norm power, ruling out the corresponding degree-only radical transfer. None of these results supplies the all-point correlation required for standard *abc*, which remains unproved here. Mathematical proofs, external analytic inputs and the narrower formal scope are distinguished.

1. THE CORRELATION PROBLEM AND THE STATUS OF THIS CONTINUATION

Author: ChatGPT. This is partial research, not a proof or a counterexample to standard *abc*. We preserve the previous distinction between an estimate valid on all triples and a construction of triples satisfying a stronger estimate. The present continuation proves two completion theorems, one elementary and one using a prime-element theorem, and a sharp ramification obstruction to one class of proposed normalizations. The completion theorems establish the required compensation on their actual outputs; they do not assert that every input triple is such an output.

Let a, b be coprime positive integers, $c = a + b$, $T = abc$, and $R = \text{rad}(T)$. Put $\zeta = (1 + \sqrt{-3})/2$, $\mathcal{E} = \mathbb{Z}[\zeta]$, $z = a + b\zeta$, and $M = N(z) = a^2 + ab + b^2$. The norm factorization and common exponent content g are as in the preceding supplements [2]. In particular, M prime implies $g = 1$; more generally $v_7(M) = 1$ implies $g = 1$.

For $j \geq 1$ define the actual arithmetic products

$$E_j(T) = \prod_{p|T} p^{(v_p(T)-j)+}, \quad C_j(T) = \prod_{p|T} p^{(j-v_p(T))+}.$$

The already established identity $TC_j(T) = R^j E_j(T)$ gives, on setting

$$I(a, b) = \log \frac{c^2}{ab}, \quad D_3(T) = \log E_3(T) - \log C_3(T),$$

$$(1) \quad I(a, b) + D_3(T) = 3 \log(c/R).$$

Indeed $E_3/C_3 = T/R^3$ and $c^2 T/(ab) = c^3$. Consequently a universal upper bound for the left side at the *abc* scale would be an equivalent reformulation, not an independently proved bridge. The new input below is instead an arithmetic construction that proves an exact formula for R on many concrete completions, including prime-norm points.

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Definition 1.1 (Specified valuation pattern and its completion). A pattern consists of a finite set S of rational primes, integers $k_p \geq 1$, and a specified arm for each $p \in S$. A completion is a primitive triple having $v_p(T) = k_p$ on the specified arm, with every other prime divisor of T having exponent one. Write

$$G = \prod_{p \in S} p^{k_p-1}.$$

Thus a completion satisfies the exact identity $T = GR$.

Lemma 1.2 (Exact compensated correlation on completions). *Every completion satisfies*

$$\frac{c}{R} = \frac{G}{ab}, \quad I(a, b) + D_3(T) = 3 \log \frac{G}{ab}.$$

In particular, $ab \geq G$ implies $c \leq R$. No norm hypothesis is required.

Proof. The exponent of each supported prime in T/R is $v_p(T) - 1$. The specified primes give G and all other primes give one. Cancel $c > 0$ in $abc = GR$, and use (1). $\square$

The point of a completion theorem is to establish such triples without assuming that their unspecified prime factors have the desired behavior.

2. AN ELEMENTARY COMPLETION THEOREM WITH MAXIMAL IMBALANCE AND CONTENT ONE

In this section let S be any finite set of primes at least five, excluding seven, and assign each specified prime to b or $c = b + 1$. Set

$$m = 1764 \prod_{p \in S} p^{k_p+1}, \quad G = \prod_{p \in S} p^{k_p-1}.$$

Choose the unique residue r modulo m specified by

$$\begin{aligned} r &\equiv 6 \pmod{36}, & r &\equiv 2 \pmod{49}, \\ r &\equiv p^{k_p} \pmod{p^{k_p+1}} & & \text{if } p \text{ is assigned to } b, \\ r &\equiv p^{k_p} - 1 \pmod{p^{k_p+1}} & & \text{if } p \text{ is assigned to } c. \end{aligned}$$

These are compatible by the Chinese remainder theorem. Empty S is allowed.

Theorem 2.1 (Uniform elementary squarefree completion). *For every real $X \geq 400m^2$, at least $X/(4m)$ integers b in the open interval $(X, 2X)$ satisfy $b \equiv r \pmod{m}$ and make $(1, b, b+1)$ a completion. Every such triple has*

$$(2) \quad v_7(b^2 + b + 1) = 1, \quad g = 1, \quad R = \frac{b(b+1)}{G}, \quad \frac{c}{R} = \frac{G}{b} < \frac{m}{X} \leq \frac{1}{20\sqrt{X}}.$$

Its imbalance is $I(1, b) = \log b + O(1)$, with an absolute constant. The assertion is uniform in the entire pattern, including its size and exponents.

Proof. The progression has at least $X/m - 1$ integers in $(X, 2X)$. The congruence modulo 36 makes the exponents of two and three in $b(b+1)$ exactly one. The congruence modulo 49 excludes seven from $b(b+1)$ and gives $b^2 + b + 1 \equiv 7 \pmod{49}$. The selected congruences give exactly k_p on the required arm; the other arm is a unit at p .

It remains to remove squares of primes $\ell \geq 5$ with $\ell \notin S \cup \{7\}$. Such a prime is coprime to m . The two conditions $b \equiv 0, -1 \pmod{\ell^2}$ give two distinct progressions modulo $m\ell^2$. Together they remove at most $2X/(m\ell^2) + 2$ integers. A square that can occur has $\ell \leq \sqrt{2X+1}$. The elementary telescoping estimate

$$\sum_{\ell \geq 5 \text{ prime}} \frac{2}{\ell^2} < 2 \sum_{n=5}^{\infty} \frac{1}{n(n-1)} = \frac{1}{2}$$

therefore leaves at least

$$\frac{X}{2m} - 1 - 2\sqrt{2X+1}.$$

For $X \geq 1$, $1 + 2\sqrt{2X+1} \leq 5\sqrt{X}$. Since $X \geq 400m^2$, we have $X/(4m) \geq 5\sqrt{X}$. The last display is consequently at least $X/(4m)$. This argument is a union bound; overlaps can only improve it. Every remaining integer has no unspecified squared prime divisor.

Now $G \leq m$ and Lemma 1.2 give the radical and ratio. The pair $(1, b)$ is primitive. Since seven is a split norm prime and $v_7(M) = 1$, the gcd of all positive split-norm exponents is one. Finally $I(1, b) = 2\log(b+1) - \log b = \log b + O(1)$. $\square$

Corollary 2.2 (Both losses can be large while their compensated sum is negative). *There are completions with $g = 1$, a large imbalance of order $\log c$, and large-prime excess of linear logarithmic size, although $c/R \rightarrow 0$. Both large arms can have prescribed exponents at least four. More precisely fix $k \geq 4$, choose primes $p < q < 2p$, $p > 7$, and prescribe exponent k on b at p and on c at q . Take $X = 400m^2$ and any completion supplied by Theorem 2.1. Then for every fixed $A > 0$, eventually $p, q > (\log c)^A$, and*

$$\frac{\log E_{3, > (\log c)^A}(T)}{\log c} \rightarrow \frac{k-3}{2(k+1)}, \quad \frac{I(1, b) + D_3(T)}{\log c} \rightarrow -\frac{3(k+3)}{2(k+1)}.$$

Proof. Primes q as stated exist by Bertrand's theorem. Here $m = 1764(pq)^{k+1}$, $G = (pq)^{k-1}$, and $\log c = 2(k+1)\log(pq) + O(1)$. The only exponents exceeding one are the two specified ones. Thus the large-prime excess is exactly $(pq)^{k-3}$ for large p . Also $\log(c/R) = \log G - \log b = -(k+3)\log(pq) + O(1)$. Divide and use Lemma 1.2. Since $\log c = O_k(\log p)$, p eventually exceeds every fixed power of $\log c$. $\square$

This is not a prime-norm family: seven divides its norm exactly once. It handles genuinely unbalanced content-one completions, not the prime-norm endpoint. The next argument treats that endpoint by a different mechanism.

3. PRIME ELEMENTS IN RECTANGLES: THE EXTERNAL INPUT AND ITS EXCEPTIONAL TERM

We use Kai's Theorem 3.1.1 [1], an extension of Mitsui's prime-element theorem retaining a possible Siegel-zero term. We state only its specialization needed here. All field constants below are absolute because $K = \mathbb{Q}(\sqrt{-3})$ is fixed.

For an integral ideal $\mathfrak{q}$, write $\Phi(\mathfrak{q}) = |(\mathcal{E}/\mathfrak{q})^\times|$. Let C be a convex open set in $|z| < 4X$, with $|z| > X$, and let α be a unit residue modulo $\mathfrak{q}$. For suitable constants $a_0, b_0 > 0$, uniformly when $N\mathfrak{q} \leq \exp(a_0\sqrt{\log X})$, the weighted prime-element count is

$$(3) \quad W(C; \mathfrak{q}, \alpha) = \frac{6}{\pi\Phi(\mathfrak{q})} \int_C (1 - \chi_{\mathfrak{q}}(z; \alpha) N(z)^{\beta_{\mathfrak{q}}-1}) d\mu(z) + O(X^2 e^{-b_0\sqrt{\log X}}).$$

Here W sums $\log N(\pi)$ over prime elements in the specified set, μ is ordinary area in $\mathbb{C}$, and the second integrand is omitted when there is no exceptional zero. Rescaling the parameter X in the source by a fixed factor changes only a_0, b_0 and the absolute implied constant. A real finite-order Hecke character is trivial on the connected infinite-place group $\mathbb{C}^\times$, so its value at a fixed residue is constant, in $\{-1, 1\}$. In particular the integrand lies between zero and two. The coefficient is $6/\pi$ because the field has six units, class number one and regulator one. In (a, b) coordinates $d\mu = (\sqrt{3}/2) da db$; we abbreviate $\kappa = 3\sqrt{3}/\pi$.

We also use the usual zero-free-region consequence for *nontrivial real finite-order* Hecke characters modulo one ideal: at most one has a real zero in

$$\beta > 1 - \frac{c_0}{\log(4N\mathfrak{q})}$$

for an absolute $c_0 > 0$; such a zero is less than one. This is the real-character part of the zero-free input recorded in [1, Theorem 2.3.1]. We do not apply a zero-free statement to the pole of the trivial character, or assert a frequency-independent assertion for arbitrary infinite-type characters. Neither GRH nor the absence of Siegel zeros is assumed. These analytic results are external inputs, not newly proved theorems or axioms in the Lean companion.

Lemma 3.1 (A single excluded conductor at a scale). *For any $Q \geq 2$, among primitive nontrivial real characters of conductor norm at most Q , at most one has a real zero*

$$\beta > 1 - \frac{c_0}{\log(4Q^2)}.$$

Call its conductor f_Q if it exists. If $N\mathfrak{q} \leq Q$ and $f_Q \nmid \mathfrak{q}$, then every exceptional term in (3) is at most $X^{-2c_0/\log(4Q^2)}$ in absolute value on C .

Proof. Two distinct primitive characters would induce distinct characters modulo the least common multiple of their conductors, whose norm is at most Q^2 . Both real zeros would be in the forbidden region for that modulus, contradicting the single-exception zero-free statement. Inducing a character changes only Euler factors, which introduce no real zeros in $(0, 1)$; primitive conductors are unique. If the single conductor does not divide $\mathfrak{q}$, any exceptional zero of that modulus is no closer to one than the displayed threshold. Since $N(z) > X^2$, the asserted decay follows. $\square$

4. PRIME-NORM COMPLETIONS WITH A MOVING VALUATION PATTERN

Now allow S to be any finite set of primes at least five, including seven, with arbitrary assignments to the three arms a, b, c . Let

$$m = 36 \prod_{p \in S} p^{k_p+1}, \quad G = \prod_{p \in S} p^{k_p-1}.$$

Choose a pair residue (a_0, b_0) modulo m by CRT: it is $(1, 6)$ modulo 36, and modulo p^{k_p+1} it is respectively

$$(p^{k_p}, 1), \quad (1, p^{k_p}), \quad (1, p^{k_p} - 1)$$

when the assigned arm is a, b, c . Each norm is a unit modulo p , and the modulo-36 norm is coprime to six. Thus $\alpha = a_0 + b_0\zeta$ is a unit modulo $m\mathcal{E}$. Exact exponents at all specified primes and exponent one at two and three are forced.

Theorem 4.1 (Uniform prime-norm squarefree completion). *There are absolute constants $\nu > 0$ and $X_0 > 1$ with the following property. Put $L = \sqrt{\log X}$ and $Q_X = e^{20\nu L}$. For $X \geq X_0$, suppose*

$$(4) \quad m \leq e^{\nu L}, \quad X e^{-\nu L} \leq U \leq X, \quad f_{Q_X} \nmid m\mathcal{E}.$$

The last condition is empty if no exceptional conductor exists. In the rectangle $U < a < 2U$, $X < b < 2X$, the sum of $\log N(a + b\zeta)$ over completions with rational-prime norm is at least

$$(5) \quad \frac{\kappa U X}{10\Phi(m\mathcal{E})}.$$

In particular such completions exist; their number is at least this quantity divided by $\log(12X^2)$. Every one has $g = 1$ and

$$(6) \quad \frac{c}{R} = \frac{G}{ab} < X^{-2} e^{2\nu\sqrt{\log X}}, \quad I(a, b) + D_3(T) < -6 \log X + 6\nu\sqrt{\log X}.$$

All constants are independent of the finite pattern, its exponents, and the position of its primes among the arms.

Proof. Uniform choice of constants. With a_0, b_0, c_0 from the preceding section, choose $\nu > 0$ sufficiently small that

$$20\nu < a_0, \quad 30\nu < b_1 := \min\{b_0, c_0/(21\nu)\}.$$

Such a choice exists. Enlarge the fixed X_0 whenever a following absolute error ratio is required to be small. Write

$$C = \{a + b\zeta : U < a < 2U, X < b < 2X\}, \quad A_X = \frac{\kappa U X}{\Phi(m\mathcal{E})}, \quad Y = e^{4\nu L}.$$

The region is convex, lies in $|z| < 4X$, and has $N(z) > X^2$. Since $\Phi(m\mathcal{E}) \leq m^2$,

$$(7) \quad A_X \geq \kappa X^2 e^{-3\nu L}.$$

Lemma 3.1 and (3) give

$$(8) \quad W(C; m\mathcal{E}, \alpha) = A_X + O(X^2 e^{-b_1 L}).$$

Indeed $2c_0 \log X / \log(4Q_X^2) \geq c_0 L / (21\nu)$ for large X . Prime elements in this strictly positive rectangle have rational-prime norm: inert prime elements are associates of rational primes and lie on the six boundary rays, not in the rectangle. The ramified prime has bounded norm and is absent for large X . A common integer divisor of a, b would have its square divide the prime norm. Thus all counted pairs are primitive.

Exact square-residue density. For $\ell \geq 5$ with $\ell \nmid m$, let $\chi_\ell = 1$ if $\ell \equiv 1 \pmod{3}$ and -1 otherwise. Then

$$\Phi(\ell^2 \mathcal{E}) = \ell^2(\ell - 1)(\ell - \chi_\ell).$$

For a split prime this is the product of two counts $\ell^2 - \ell$; for an inert prime it is $\ell^4 - \ell^2$. Among unit pair residues modulo ℓ^2 , the condition $\ell^2 \mid a$ has $\ell^2 - \ell$ possibilities for b . The same count holds for b and $a + b$. These three sets are disjoint among units, since an intersection makes both coordinates zero modulo ℓ . Their combined relative density is therefore

$$(9) \quad \beta_\ell = \frac{3}{\ell(\ell - \chi_\ell)}.$$

CRT combines each bad residue with the fixed residue modulo m . For large X we have $Y < X$; the rational prime norm is greater than X^2 , so every candidate prime element is a unit at each $\ell \leq Y$. The unit restriction in this count therefore omits no small-square candidate.

Small square divisors, including exceptional characters. For $\ell \leq Y$, the refined modulus $m\ell^2 \mathcal{E}$ has norm at most $e^{18\nu L} < Q_X$. We do not assume these refined moduli are nonexceptional. The integrand bound by two in (3) shows that their bad residue classes have total weighted count at most

$$2A_X \beta_\ell + O(\ell^2 X^2 e^{-b_0 L}).$$

The union bound over primes up to Y gives a loss at most

$$(10) \quad 2A_X \sum_{\ell \geq 5 \text{ prime}} \beta_\ell + O(X^2 Y^3 e^{-b_1 L}).$$

The last error divided by (7) is $O(e^{-(b_1 - 15\nu)L}) = o(1)$ uniformly. There is an explicit positive margin despite the factor two:

$$2 \sum_{\ell \geq 5 \text{ prime}} \beta_\ell \leq 6 \sum_{j=2}^{\infty} \frac{1}{2j(2j+1)} = 5 - 6 \log 2 < \frac{23}{27}.$$

For completeness, partial fractions give $\sum_{j \geq 1} (2j(2j+1))^{-1} = 1 - \log 2$, and removing $j = 1$ gives the equality. Moreover $\log 2 = 2 \sum_{j \geq 0} ((2j+1)3^{2j+1})^{-1} > 56/81$, which proves the last strict inequality. Hence at least $4/27$ of the main term remains before the uniform errors and the large-square tail.

Large square divisors. If $\ell > Y$ and ℓ^2 divides one of the three positive arms, then $\ell \leq 2\sqrt{X}$. Ignoring primality and the modulo- m condition, for each arm the number of such pairs is $O(X^2/\ell^2 + X)$. Summing over integers in the allowed prime range gives

$$O(X^2/Y + X^{3/2}).$$

Each pair has logarithmic norm weight at most $\log(12X^2)$. Relative to (7), this loss is

$$O(e^{-\nu L} \log X + e^{3\nu L} X^{-1/2} \log X) = o(1).$$

Primitivity ensures that a square in T occurs in just one arm. After subtracting both tails from (8), the remaining weight is at least $A_X/10$ for sufficiently large X , uniformly in all data. Specified primes and two and three were already treated by congruences; thus every remaining point is a completion.

Finally $G \leq m \leq e^{\nu L}$ and $ab > UX \geq X^2 e^{-\nu L}$. Apply Lemma 1.2 to obtain (6). Prime norm supplies $g = 1$, with no compressed exponent used. $\square$

Corollary 4.2 (Arbitrary specified depths also at two and three). *In Theorem 4.1, the fixed exponents one at two and three may be replaced by any positive integers k_2, k_3 , on any assigned arms. Replace the factor 36 in m by $2^{k_2+1}3^{k_3+1}$ and multiply G by $2^{k_2-1}3^{k_3-1}$. The same absolute constants and conclusions hold under the resulting modulus condition. In Theorem 2.1 the same replacement is possible, with assignments only to b, c and the factor 49 retained.*

Proof. At $p = 2, 3$, use the same three pair residues $(p^{k_p}, 1), (1, p^{k_p}), (1, p^{k_p} - 1)$ as for other primes. Their norms are one modulo p , and they force the required exact valuation on one arm with the other two arms units. All small-square sieving still starts at five, and all estimates use only m and G . In the elementary case select $b \equiv p^{k_p}$ or $p^{k_p} - 1$; the same union bound excludes all other squares. No analytic constant or square-density margin changes. $\square$

5. CONSEQUENCES, GROWTH OF THE PRESCRIBED DEPTH, AND LIMITATIONS

Corollary 5.1 (Every fixed pattern has infinitely many prime-norm completions). *For each fixed finite pattern of Definition 1.1, with primes at least five and arbitrary arm assignments, infinitely many completions have prime auxiliary norm. They can be taken in $X < a, b < 2X$, and all sufficiently large such completions produced here satisfy $c < R$.*

Proof. For fixed m , there are finitely many real characters of conductor dividing $m\mathcal{E}$. None has a zero at one. Thus their real zeros eventually lie outside the shrinking exceptional interval in Lemma 3.1. Condition (4) holds with $U = X$ for all sufficiently large X . The positive lower bound in Theorem 4.1 proves existence at arbitrarily large height. This corollary needs no assertion that Siegel zeros never occur, and makes no uniform claim about its fixed-pattern threshold. $\square$

Corollary 5.2 (Growing large primes and growing depth at prime norm). *There is a sequence of prime-norm completions with $g = 1$, a supported prime p exceeding every fixed power of $\log c$, and $v_p(T) \rightarrow \infty$. Only this prime has exponent exceeding one, and $R = c^{3-o(1)}$ on the sequence.*

Proof. Use the absolute ν of Theorem 4.1. Let $k = \lfloor (\log X)^{1/4} \rfloor$ and $Z = \exp(\nu\sqrt{\log X}/(2(k+1)))$. Bertrand's theorem gives two distinct primes in $(Z, 4Z)$ for large X . For either prime p , $m_p = 36p^{k+1} \leq e^{\nu\sqrt{\log X}}$, since $\log 36 + (k+1)\log 4 = o(\sqrt{\log X})$. The exceptional conductor cannot divide $36\mathcal{E}$ for arbitrarily large X , by the fixed finite-character argument of Corollary 5.1. It can divide at most one of the two ideals $m_p\mathcal{E}$: otherwise it would divide their gcd, $36\mathcal{E}$. Choose the other prime and prescribe exponent k on, for example, c . The theorem with $U = X$ constructs a completion. We have $2X < c < 4X$, $k \rightarrow \infty$, and $\log p \asymp (\log X)^{1/4}$, so $p > (\log c)^A$ for every fixed A eventually. Finally $G = p^{k-1} = \exp(O(\sqrt{\log X}))$, while $T \asymp X^3$. The identity $R = T/G$ proves $R = c^{3-o(1)}$. $\square$

Boundary of the conclusion. The last corollary is an existence assertion about exactly compensated completions, not a bound on every prime-norm triple. In its construction $\log G = O(\sqrt{\log c})$, rather than a positive fixed fraction of $\log c$. The earlier ray-class obstruction allows prescribed prime powers with a much smaller height budget; the present prime-element error term does not give a squarefree completion uniformly at that polynomial-modulus scale. Similarly, Theorem 4.1 allows the aspect ratio U/X to shrink only as $e^{-\nu\sqrt{\log X}}$, not as a fixed negative power of X . Theorem 2.1 handles the very unbalanced case $a = 1$ but ensures only content one, not prime norm. These limitations are part of the theorems, not omitted exceptional cases.

6. A SHARP RAMIFICATION COST OF FORCING HIGH NORM EXPONENT

A separate proposal is to move arbitrary $g = 1$ triples to large- g triples by a rational map, then invoke the previously proved high-content bounds. The following theorem measures an unavoidable geometric cost of maps which force a norm power identically. It does not address arbitrary point-adaptive maps or arithmetic cancellations among their values.

Let $A(X, Y), B(X, Y) \in \mathbb{Q}[X, Y]$ be homogeneous forms of the same degree $d \geq 1$, having no common zero on $\mathbb{P}^1(\mathbb{C})$. They define a nonconstant morphism $f = [A : B]$ of degree d . Let

$$F = AB(A + B), \quad s = \#\{P \in \mathbb{P}^1(\mathbb{C}) : F(P) = 0\}.$$

Thus s is the homogeneous degree of the geometric squarefree boundary. All root counts include the point at infinity.

Theorem 6.1 (Norm-power ramification budget). *Suppose*

$$A^2 + AB + B^2 = UV^g, \quad g \geq 2,$$

where U, V are nonzero homogeneous forms, $\deg U = h$, $\deg V = v$, so $h + gv = 2d$. The factors need not be squarefree or coprime. Then

$$(11) \quad s \geq d + 2 + (g - 1)v = d + 2 + (1 - 1/g)(2d - h).$$

In particular, for a pure power $A^2 + AB + B^2 = \kappa Q^g$ with $\kappa \neq 0$, one has $g \mid d$ and

$$(12) \quad s \geq 3d - 2d/g + 2.$$

Proof. We give the required ramification count directly. The homogeneous Jacobian

$$J = A_X B_Y - A_Y B_X$$

is a nonzero form of degree $2d - 2$. Nonvanishing follows from nonconstancy in characteristic zero. At any source point choose source and target affine coordinates such that the latter is $f(t) = f(0) + ut^e + \dots$, $u \neq 0$. The order of the Jacobian there is $e - 1$, since its local expression differs from $f'(t)$ by an invertible factor; Euler's identity accounts for the homogeneous coordinates. This also applies in the charts at infinity. Consequently the sum of $e_P - 1$ over all source points is $2d - 2$.

The three fibres over $0, -1, \infty$ are disjoint and have total multiplicity $3d$. Their contribution to this ramification sum is exactly $3d - s$. The two zeros of $t^2 + t + 1$ are distinct from these three values. Their pullback, $A^2 + AB + B^2$, has total multiplicity $2d$ and at most $h + v$ distinct zeros, since its support is contained in that of UV . Its ramification contribution is at least $2d - h - v = (g - 1)v$. All other ramification contributions are nonnegative. Hence

$$3d - s + (g - 1)v \leq 2d - 2,$$

proving (11). For a pure power, the two forms $A - \rho B$ and $A - \bar{\rho} B$, with $\rho, \bar{\rho}$ the roots of $t^2 + t + 1$, have disjoint supports and each has degree d . Every root multiplicity is divisible by g ; thus $g \mid d$. Substitute $h = 0$ in the general bound. $\square$

Corollary 6.2 (No boundary-Belyi exact-power normalization). *A nonconstant map branched only over $\{0, -1, \infty\}$ cannot satisfy $A^2 + AB + B^2 = UV^g$ with $g \geq 2$ and $\deg V > 0$.*

Proof. Both norm-root fibres lie outside these three branch values and must be unramified. A zero of V would have multiplicity at least g in their pullback and hence would be ramified. This is a contradiction. The assertion concerns this specific norm-power normalization, not every use of Belyi maps in an arithmetic height argument. $\square$

Corollary 6.3 (Limit of a degree-only radical transfer). *In the pure-power case, assume the three original boundary points $XY(X + Y) = 0$ belong to the support of F . If $D = s - 3$ is the degree of the remaining squarefree boundary, then $D \geq d + 1$. Consequently a transfer that only bounds the new boundary radical by its polynomial height degree cannot prove an abc height bound for the input from an abc bound for the output.*

Proof. Write $d = gm$, $m \geq 1$. Equation (12) gives $D \geq 3d - 2m - 1$. Subtracting $d + 1$ leaves $2(g - 1)m - 2 \geq 0$. To specify the last assertion precisely, the degree-only estimates for a fixed map have the shape $H(f(x)) \gg H(x)^d$ and $\text{rad}(F(x)) \ll R(x)H(x)^D$, up to constants for fixed coefficients and denominators. Combining these with an output bound of exponent $1 + \epsilon$ gives only

$$H(x)^{d - (1 + \epsilon)D} \ll R(x)^{1 + \epsilon}.$$

The exponent of $H(x)$ is negative. This does not bound $H(x)$ above. This is an obstruction to that expressly stated transfer mechanism, not to sharper arithmetic bounds or to all rational transformations. $\square$

Proposition 6.4 (Sharpness and the quadratic example). *For every $g \geq 2$, write*

$$(X + Y\zeta)^g = A_g(X, Y) + B_g(X, Y)\zeta.$$

Then A_g, B_g have integer coefficients, define a degree- g morphism, satisfy $A_g^2 + A_g B_g + B_g^2 = (X^2 + XY + Y^2)^g$, and their boundary has exactly $3g$ distinct geometric zeros. Equality holds in (12). For $g = 2$ the precise factorization is

$$A_2 = X^2 - Y^2, \quad B_2 = 2XY + Y^2, \\ A_2 B_2 (A_2 + B_2) = XY(X + Y)(X - Y)(2X + Y)(X + 2Y).$$

Proof. Norm multiplicativity proves the norm identity. Over $\mathbb{C}$,

$$\frac{A_g + B_g \zeta}{A_g + B_g \bar{\zeta}} = \left(\frac{X + Y\zeta}{X + Y\bar{\zeta}} \right)^g.$$

The two linear forms in the right side have distinct zeros. Thus the map is conjugate to $t \mapsto t^g$ and has no common zero in its two coordinates. It ramifies only above the two norm roots, neither a boundary target. Each of the three boundary targets therefore has g distinct preimages. The three original boundary points are among them because units remain units after raising to a power. Direct expansion gives the quadratic formulas. $\square$

7. CONCRETE CERTIFICATES AND A SAME-HEIGHT LIMITATION

A prime-norm completion with a fourth power on one arm is

$$(a, b, c) = (225001, 228750, 453751), \quad M = 154420991251 \quad \text{prime}.$$

Its complete arm factorizations are

$$a = 7 \cdot 32143, \quad b = 2 \cdot 3 \cdot 5^4 \cdot 61, \quad c = 587 \cdot 773.$$

Thus $G = 125$, $g = 1$, and the exact ratio is $c/R = 125/(225001 \cdot 228750)$. The verifier also certifies a first fourth power at eleven and a sixth power at five in two further prime-norm completions; these are finite examples, not tests of the unknown analytic threshold in Theorem 4.1.

An unbalanced example with both large arms having fourth-power factors is

$$1 + 27960078750 = 27960078751, \\ b = 2 \cdot 3 \cdot 5^4 \cdot 7456021, \quad c = 11^4 \cdot 887 \cdot 2153.$$

Its norm is seven modulo 49, so it has content one. Its only repeated primes are five and eleven, and $c/R = 5^3 11^3/b$ exactly. All displayed factor primes and all three asserted prime norms are certified with a recursive full- $n - 1$ Lucas test. At a node n , the factorization of $n - 1$ is checked against previously certified smaller primes; a base a satisfies $a^{n-1} \equiv 1 \pmod{n}$ and $\gcd(a^{(n-1)/q} - 1, n) = 1$ for every $q \mid n - 1$. For every prime divisor ℓ of n , these conditions force $n - 1 \mid \ell - 1$, hence $\ell = n$. This proves the criterion and validates its recursive use without a probable-prime assumption.

The replay also implements the entire elementary sieve in an actual range satisfying its uniform threshold. With S empty, $m = 1764$, $r = 1374$ and $X = 1244678400 = 400m^2$, there are 705600 progression points. Sieving all 5120 relevant prime-square bases leaves 610293 completions, compared with the proved lower bound 176400. One hundred surviving pairs are independently factored to cross-check the progression-index implementation. Exhaustive local counts examine 260070 pairs modulo p^2 for $p = 5, 7, 11, 13, 17, 19$. A separate mod- p lift calculation agrees with their unit counts. Twelve exact polynomial computations verify the pure-power maps and the geometric boundary degrees, including the point at infinity. The prime certificates have 50 nodes and 195 modular conditions.

A limitation on using completions to bound an arbitrary input is worth making quantitative. If a triple were to satisfy $c > KR^{1+\epsilon}$, then its full repeated-prime factor would obey

$$\frac{T}{R} > \frac{K^{1/(1+\epsilon)}}{2} c^{1+\epsilon/(1+\epsilon)},$$

because $T \geq c^2/2$. Any modulus prescribing all these repeated valuations is at least T/R . At a comparable height $X \asymp c$, this is outside both the modulus range $m \leq \sqrt{X}/20$ in the elementary theorem and the range $m \leq e^{\nu\sqrt{\log X}}$ in the prime-norm theorem. A completion at a much larger height does not preserve the original triple's radical defect. Thus the counting results cannot be silently substituted for an all-point comparison theorem. A new argument would have to recover missing correlation in precisely this sparse, large-modulus regime.

8. PROOF DEPENDENCIES, COMPUTATIONS, AND THE UNRESOLVED ALL-POINT ESTIMATE

The elementary chain is CRT plus a union bound for squares of two consecutive linear forms. It produces genuine content-one triples, with explicit polynomial height cost and exact radical compensation, at maximal imbalance. The prime-norm chain uses two expressly stated established analytic inputs, then proves a local density, a uniform exceptional-conductor lemma, and small- and large-square estimates. It constructs completions with variable patterns and rational-prime norm. The geometric chain is a direct Jacobian proof of ramification counting followed by two disjoint pullback calculations. None uses a conjectural height inequality.

The three chains do not establish the all-point estimate

$$\forall \epsilon > 0 \exists K_\epsilon > 0 \forall (a, b, c), \quad c \leq K_\epsilon \text{rad}(abc)^{1+\epsilon}.$$

In particular the positive count of completions does not exclude an infinite sparse family outside the completion conditions. Siegel zeros were not assumed absent: the moving-pattern theorem states its excluded conductor, and the growing-prime corollary proves how to avoid it. No numerical replay can evaluate the unknown absolute analytic constants or certify a uniform large-height threshold.

The finite standard-library verifier records exact local residue counts, actual prime-norm completions, content-one unbalanced examples, independently factored radicals, and polynomial identities. Counts and hashes are recorded only after execution. A separate scoped Lean companion proves elementary integer identities, exact fixed congruence consequences, finite weighted-union accounting, explicit compensation implications, and the numerical degree-budget consequence. It does not formalize Kai's theorem, Hecke characters, the infinite sieve argument, or the general Jacobian ramification theorem. It must not be described as a formal proof of standard abc or of this entire paper. Earlier uncompiled compression source remains explicitly uncompiled unless its own exact file is separately checked.

Direction	Result in this continuation	Remaining issue
Elementary / imbalance	Uniform squarefree completion; exact $g = 1$ and compensation	Does not cover arbitrary input triples or require prime norm
Prime elements / sieve	Moving-pattern prime-norm completion; high primes and depth can grow	Exceptional conductor stated; polynomial-modulus and very thin regions uncontrolled
Rational normalization	Sharp norm-power ramification cost	Does not forbid arithmetic cancellations or adaptive non-power constructions
Global correlation	Exact identity (1), not asserted new	Required uniform bound is still at abc strength
Other manuscript routes	Their established inputs are retained	IUT all-place comparison, FCRT uniform residual, and general Pell/Mersenne depth gates remain open

The completion method is a new derivation in this project, not a claim of literature priority. Independent autonomous agents and external peer review are not asserted. The correct next all-point problem is to bound the uncontrolled complementary configurations, rather than to rename (1) or to infer a universal statement from positive density.

REFERENCES

- [1] W. Kai, *Notes on Mitsui's Prime Number Theorem with Siegel zeros*, arXiv:2209.11816v2, July 26, 2023. Theorem 3.1.1 and the real finite-order character part of the zero-free input in Section 2.3. [Author preprint](#).
- [2] ChatGPT, *Uniformity, Prime Support, and Reachable Lattices in Approaches to the abc Conjecture*, repository [wyx66624/ABCConjecture](#), main snapshot [937cb77058a80e9e9ae8ce3a4c8e13b752119796](#); supplemented by the attached exponent-compression source of September 6, 2026. These are partial-research sources, not established proofs of *abc*.